R&D | RetailPulse

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Data Dictionary and Data Model

> **In simple words**
>
> This document is a map of our data: what each column means, how tables join, and the traps to avoid (holiday double counting, missing dates, stores not open yet).

1. Tables we use

Table	Grain (one row is...)	Use
train	one store, one family, one date	Actual sales (main fact)
stores	one store	City, state, type, cluster
holidays_events	one holiday or event record	Holiday and event flags
oil	one date	Optional context
transactions	one store, one date	Optional context

We do **not** use `test.csv`. It has no sales, so we cannot measure error on it.

2. Column meanings

train

Column	Meaning	Note
id	Row ID	
date	Sales date	Daily data. Some dates are missing (for example 25 December)
store_nbr	Store ID	Links to stores
family	Product category	Our category field
sales	Units sold	Can have decimals (items sold by weight)
onpromotion	Items on promotion that day	A count, not only 0 or 1

stores

Column	Meaning	Example
store_nbr	Store ID	1, 2, 3
city	City	Quito
state	State	Pichincha
type	Store type	A, B, C, D, E
cluster	Group of similar stores	1 to 17

holidays_events

Column	Meaning	Values
date	Date of the record	
type	Kind of record	Holiday, Event, Transfer, Bridge, Additional, Work Day
locale	Scope	National, Regional, Local
locale_name	Region or city name	Ecuador, Pichincha, Quito
description	Name	Christmas
transferred	Holiday moved to another day	True / False

3. Things to know before you start

- Public-sector wages are paid on the **15th and the last day** of each month. This can change sales (payday effect).
- A big earthquake hit Ecuador in **April 2016**. It appears in the events data. Treat it as a special event.
- Some stores opened late. Some families have zero sales for long periods. Both can distort the baseline.
- After loading, check counts with the Kaggle page: about 3 million rows in train, **54 stores**, **33 families**.

4. Holiday join: how to avoid double counting

> **The risk**
>
> One date can have more than one holiday record. Joining train to holidays_events on `date` alone repeats sales rows and **double counts sales**.

1. Remove rows where `transferred = True`. That date was not really celebrated.
2. Match by scope: **National** = all stores. **Regional** = `stores.state = locale_name`. **Local** = `stores.city = locale_name`.
3. Build a **store-date holiday flag table** with exactly one row per `store_nbr` and `date`: columns store_nbr, date, is_holiday. Use DISTINCT or EXISTS.
4. Join train to this flag table, never to holidays_events directly.

is_holiday = 1 only for types Holiday, Transfer, Additional and Bridge. **Event** is kept separate (for example Black Friday). **Work Day** is a normal day.

5. Grain of the error table

Grain means what one row stands for. Our error table has one row per `store_nbr` x `family` x `date`. We add flags to it: is_holiday, is_payday, onpromotion, month, quarter, season. All WAPE numbers come from summing these rows.

6. Data model for Power BI (star schema)

```
dim_date      dim_store      dim_family
    \            |            /
     \           |           /
          fact_error
  (one row per store x family x date)
  flags inside the fact: is_holiday, is_payday
```

Fact columns: date, store_nbr, family, actual_sales, forecast_sales, abs_error, excess_units, shortfall_units, onpromotion, is_holiday, is_payday. Dimensions give names and groups for slicers.

7. Data cleaning rules (done in SQL)

Check	Action
NULL in sales or date	Count them. Remove or explain.
Duplicate rows (same store, family, date)	Check count. Remove if found.
Missing dates	Expected for 25 December. Do not fill. Make sure baseline does not use a wrong day.
Rows before a store's first sale date	Exclude. The store may not be open yet.
Families with zero sales for long periods	Mark them. Exclude or flag in high-error rule.

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